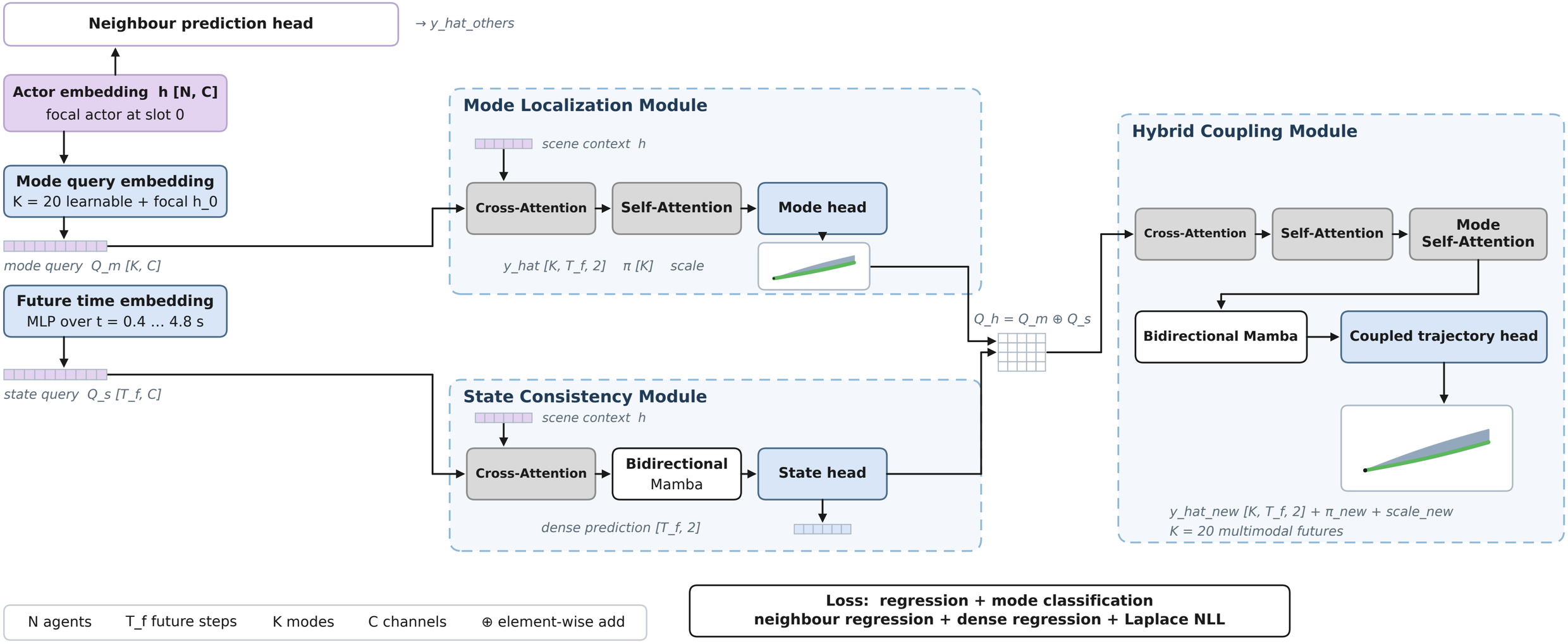
Neighbour prediction head
→ y_hat_others
Actor embedding h [N, C]
focal actor at slot 0
Mode query embedding
K = 20 learnable + focal h_0
mode query Q_m [K, C]
Future time embedding
MLP over t = 0.4 … 4.8 s
state query Q_s [T_f, C]
Mode Localization Module
scene context h
Cross-Attention
Self-Attention
Mode head
y_hat [K, T_f, 2] π [K] scale
State Consistency Module
scene context h
Cross-Attention
Bidirectional Mamba
State head
dense prediction [T_f, 2]
Q_h = Q_m ⊕ Q_s
Hybrid Coupling Module
Cross-Attention
Self-Attention
Mode Self-Attention
Bidirectional Mamba
Coupled trajectory head
y_hat_new [K, T_f, 2] + π_new + scale_new
K = 20 multimodal futures
N agents
T_f future steps
K modes
C channels
⊕ element-wise add
Loss: regression + mode classification
neighbour regression + dense regression + Laplace NLL